

# Software Requirements Specification (SRS)

**Project:** Samartha Ayurveda NFC Marketing & Gamification Platform

**Version:** 2.0 (Enterprise Edition)

## 1. Project Overview

A centralized web application designed for the Samartha Ayurveda publicity team. The system uses NFC technology for physical marketing and includes a gamified employee tracking dashboard to monitor, motivate, and reward the sales force.

## 2. User Roles & Permissions

| Role                  | Permissions                                                                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Admin (Owner/Manager) | Full system control. Can assign/pause NFC cards, set daily global quests, send notices, view all employee stats, manage rewards, and export database records to Excel. |
| User (Employee)       | Access to personal Profile (Digital ID), Daily Goals tracker, Inbox (Admin Notices), and the Weekly Leaderboard. Cannot edit ad destinations.                          |

## 3. Functional Requirements

### 3.1. Admin Panel (The Command Center)

- **Card & Employee Management:** Link physical NFC cards to employee profiles. Ability to instantly "Pause" or "Terminate" an NFC card's redirect link.
- **Gamification Engine (Quests & Rewards):**
  - Set daily/weekly scan targets (e.g., "Get 15 unique taps today").
  - Define point values or tangible rewards for hitting milestones.
- **Notice Board System:** Broadcast important messages, updates, or motivational alerts to all employees or specific individuals.
- **Data Export (Excel/CSV):** A single-click button to download tap logs, employee performance metrics, and lead generation data. *(Technical tip: If using Python Flask, the `pandas` or `Flask-Excel` libraries make this export feature very easy to implement).*

### 3.2. Employee Panel (The Motivation Dashboard)

- **Digital Profile (ID Card View):** A dedicated tab displaying their official Samartha Ayurveda credentials (Photo, Employee ID, Contact Info, Address).
- **Daily Goals & Progress:** A visual progress bar showing their current unique taps vs. the daily goal set by the admin.
- **Weekly Leaderboard:** A ranking table showing the top-performing employees of the week based on unique interactions, fostering healthy competition.
- **Inbox/Notifications Tab:** A feed where they can read the notices and updates sent by the Admin.

---

## 4. Database Schema (Logical Design)

To support the ID card features and the gamification, the database requires an expanded structure.

**Table 1: Employees (The Digital ID Card)**

| Field     | Type        | Description                          |
|-----------|-------------|--------------------------------------|
| emp_id    | Primary Key | Unique company ID (e.g., SA-001).    |
| name      | String      | Full legal name.                     |
| email     | String      | Official contact email.              |
| phone     | String      | Contact number.                      |
| address   | Text        | Residential or branch address.       |
| photo_url | String      | Path/URL to the employee's headshot. |

|              |         |                             |
|--------------|---------|-----------------------------|
| total_points | Integer | Gamification score tracker. |
|--------------|---------|-----------------------------|

Table 2: NFC\_Cards

| Field           | Type            | Description                      |
|-----------------|-----------------|----------------------------------|
| card_id         | Primary Key     | Unique ID for the physical chip. |
| assigned_emp_id | Foreign Key     | Links to Employees.emp_id.       |
| slug            | String (Unique) | The URL path (e.g., "/t/emp01"). |
| status          | Enum            | Active, Paused, Terminated.      |

Table 3: Tap\_Logs (Analytics)

| Field      | Type        | Description                    |
|------------|-------------|--------------------------------|
| log_id     | Primary Key | Unique event ID.               |
| card_id    | Foreign Key | Which card was tapped.         |
| ip_address | String      | To filter out duplicate taps.  |
| timestamp  | DateTime    | When the interaction happened. |

Table 4: Daily\_Goals

| Field        | Type        | Description                                   |
|--------------|-------------|-----------------------------------------------|
| goal_id      | Primary Key | Unique goal ID.                               |
| date         | Date        | The specific day the goal applies to.         |
| target_scans | Integer     | How many unique taps are required.            |
| reward_desc  | String      | What the employee gets (e.g., "Bonus +\$10"). |

**Table 5: Notifications**

| Field         | Type        | Description                        |
|---------------|-------------|------------------------------------|
| notice_id     | Primary Key | Unique message ID.                 |
| target_emp_id | Foreign Key | Set to "ALL" or a specific emp_id. |
| message       | Text        | The content of the admin's notice. |
| timestamp     | DateTime    | When it was sent.                  |

## 5. System Workflow (Example Scenario)

1. **Morning Setup:** The Admin logs in, sets the Daily Goal to "20 Unique Reaches," and sends a Notification: *"Happy Monday Team! First to 20 gets a coffee voucher."*
2. **Field Work:** Employee Rahul checks his Employee Panel on his phone. He sees his Profile, reads the notice, and sees his progress bar is at 0/20.
3. **The Interaction:** Rahul taps his NFC card on a client's phone. The backend logs the client's IP, checks if the IP is new for today, and registers a valid tap.

4. **Live Update:** Rahul's progress bar instantly jumps to 1/20. The Weekly Leaderboard updates, pushing Rahul to 1st place.
  5. **End of Month:** The Admin clicks the **"Export Data"** button to download an Excel sheet of all logs to calculate bonuses based on the total\_points accumulated.
-